Deliverable 2
Structural Aliasing Analysis in Time-Series Forecasting Using Chronos

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This deliverable reports how the plan approved in Deliverable 1 was carried out. The presentation proceeds from the trained models to the frequency behaviour measured on them, then to the Bayesian analysis that tests it, the semantic layer that operationalises it, the proposed mitigation strategy, the results, and finally the security reading.

Two terms are used throughout in a precise sense. Phase-locking is a property of the input signal and the tokenisation geometry alone, exact arithmetic on (f, f_s, P, S) that holds for any model, trained or not; Deliverable 1 derives it as the two conditions $x[n+S] = x[n]$ and $x[n+P] = x[n]$. A frequency satisfying the first is a stride lock, one satisfying the second a patch null, and the sets they generate are the stride branch and the patch branch of $\mathcal{F}_{lock}$, of which $\mathcal{F}_{lock}$ is the union. Structural aliasing, by contrast, is an empirical property of the learned representations, in which distinct inputs are mapped to nearly identical latent states. Phase-locking is a necessary but not a sufficient condition for it; the distinction is developed in the final deliverable.

Frequency Analysis and Data Observations

The first analysis sweeps pure sinusoids from 2 to 250 Hz in one-hertz steps, at a sampling frequency of $f_s = 512$ Hz. For each tone, the frequency produced by Chronos is compared with the known input frequency.

P	S	O	**tokens**	$S \mid P$	$P \mid S$
8	8	0.000	256	yes	yes
16	8	0.500	255	yes	**no**
16	12	0.250	170	**no**	**no**
16	16	0.000	128	yes	yes
24	24	0.000	85	yes	yes

Table 1: *The five geometries analysed here, carried over from Deliverable 1. The token count is the number of patch tokens at the* 2048*-sample training context. The last two columns are divisibility conditions, and each states that one branch is contained in the other:* $S \mid P$ *means the patch holds a whole number of strides, so every stride lock is also a patch null, and* $P \mid S$ *the converse, that every patch null is also a stride lock. Both hold only at* $P = S$*, where the two branches are the same set of frequencies.*

During the preparation of this work, the authors used AI-based tools, including GPT-5.5, Gemini 3.6 Flash, Sonnet 5, and Opus 4.6, to support language refinement and improve clarity. All generated or suggested content was carefully reviewed, revised where necessary, and validated by the authors, who assume full responsibility for the final content of this work.

Model	f_s/S [Hz]	f_s/P [Hz]	**stride only**	**both**	**patch only**
`p8-s8`	64	64	—	64 128 192	—
`p16-s8`	64	32	—	64 128 192	32 96 160 224
`p16-s12`	42.67	32	42.67 85.33 170.67 213.33	128	32 64 96 160 192 224
`p16-s16`	32	32	—	32 64 96 128 160 192 224	—
`p24-s24`	21.33	21.33	—	21.33 42.67 64 85.33 106.67 128 149.33 170.67 192 213.33 234.67	—

Table 2: *The locked frequencies of the five geometries carried over from Deliverable 1, with each frequency placed under the branch that produces it. A geometry with empty outer columns is one where $S \mid P$: the two branches coincide everywhere and no frequency is attributable to either alone.*

Pure sinusoids keep the test easy to interpret: no background spectral content can explain a loss at a predicted lock frequency.

Table 1 summarises the patch size P, stride S, overlap $O = (P - S)/P$, and token count of each configuration. All models, including the $P{=}S{=}16$ baseline, are trained from scratch with the same 100k-step budget and seed; the published `chronos-bolt-tiny` checkpoint is not used. This keeps training length fixed across configurations, while the token count shows the computational cost of overlap.

A locked frequency can satisfy the stride condition, the patch condition, or both. The notation $S \mid P$ means that P is an integer multiple of S. In that case every stride-lock site is also a patch-null site, so there are no stride-only sites, although patch-only sites may remain. Conversely, $P \mid S$ means that every patch-null site is also a stride-lock site, so there are no patch-only sites. If neither relation holds, both types of exclusive site are possible; if $P = S$, the two sets coincide.

Table 2 applies this rule to the sweep by sorting each candidate frequency into stride only, both, or patch only. Its main purpose is attribution: a dip at an exclusive site can be associated with one branch, whereas a dip at a shared site cannot separate their contributions.

Each configuration is swept over the same band at several non-redundant starting phases. The nominal 512-sample context is shortened, when necessary, to a whole number of strides so that the final patch is not padded; otherwise padding could create an artificial collapse. When the stride divides 512, integer-frequency tones also complete whole cycles in the window. For other strides, the context differs by fewer than S samples. Each model is loaded, measured, and released in turn.

Observations

What follows are the observations read off the sweep of the five geometries of Deliverable 1.

Frequency Reconstruction This reading is behavioural and records what the model outputs. The sinusoid is given as context, the median forecast is appended and fed back, and the rollout continues to 512 generated samples. Its dominant frequency is read as the largest magnitude of the discrete Fourier transform after the mean is removed and the zero-frequency bin skipped, then averaged over phases and plotted against the input frequency. A point on the diagonal is a tone rebuilt correctly, a flat segment is a set of distinct inputs collapsing onto one output, and the shaded band is one standard deviation over phases.

Taking into account the frequency of the input tone, the dominant frequency of the rollout is plotted against it in Figure 1. The dotted diagonal is a perfect reconstruction and the dashed verticals are $\mathcal{F}_{lock}$.

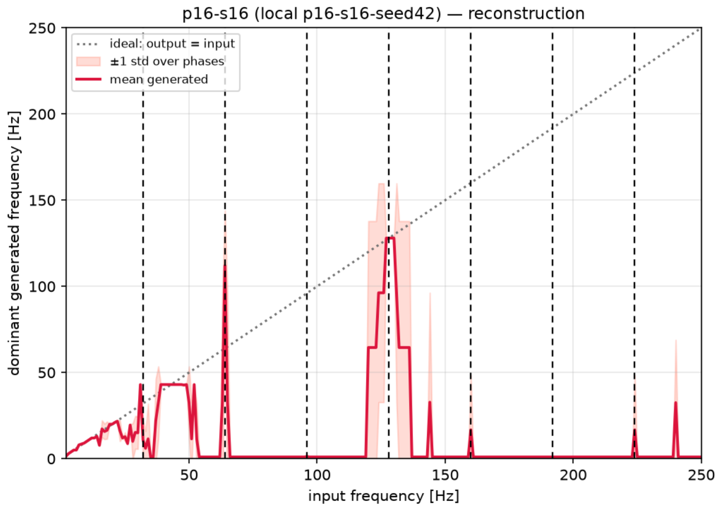

Figure 1: *Reconstruction on the $P = 16, S = 16$ geometry: dominant frequency of the rollout against the tone supplied, averaged over phases, one standard deviation shaded. The dotted diagonal is a tone rebuilt correctly; the dashed verticals are $\mathcal{F}_{lock}$.*

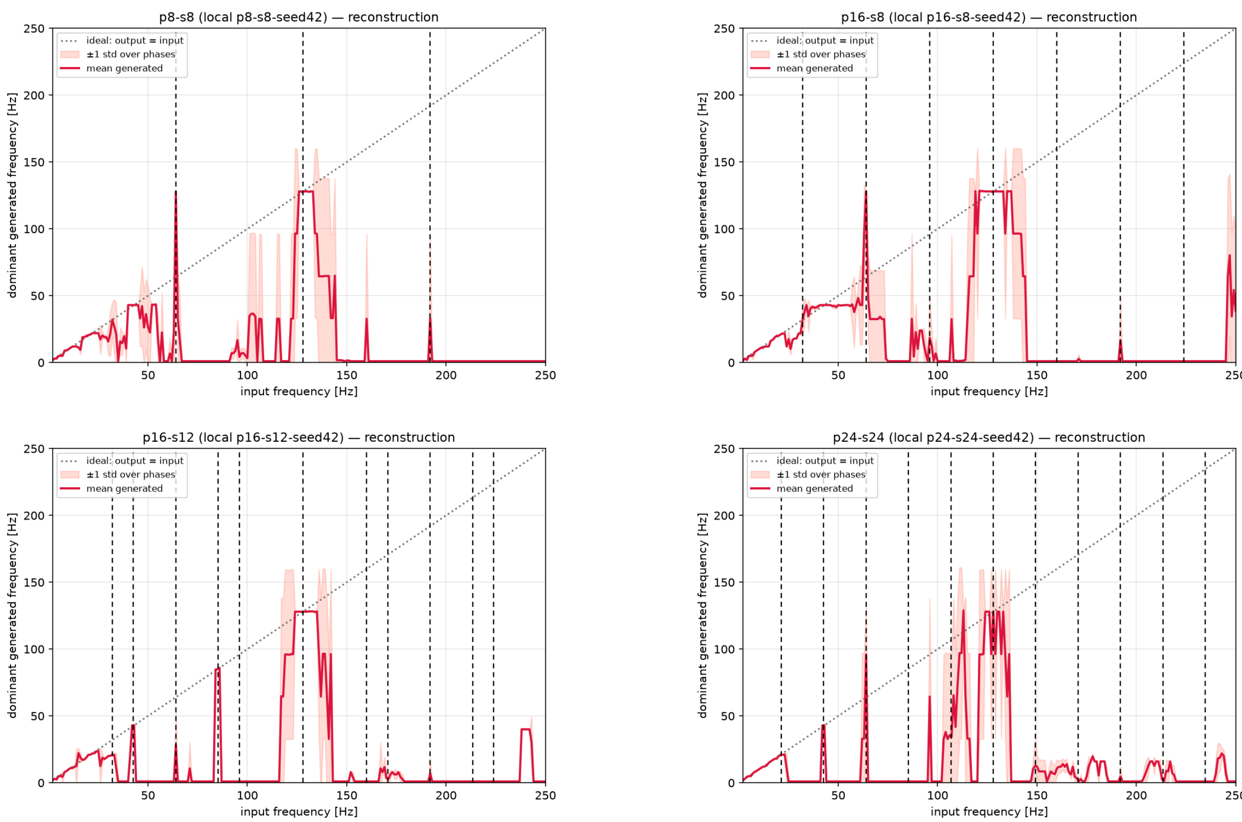

Figure 2: *Reconstruction on the other four geometries: dominant frequency of the rollout against the tone supplied, averaged over phases, one standard deviation shaded. The dotted diagonal is a tone rebuilt correctly; the dashed verticals are $\mathcal{F}_{lock}$.*

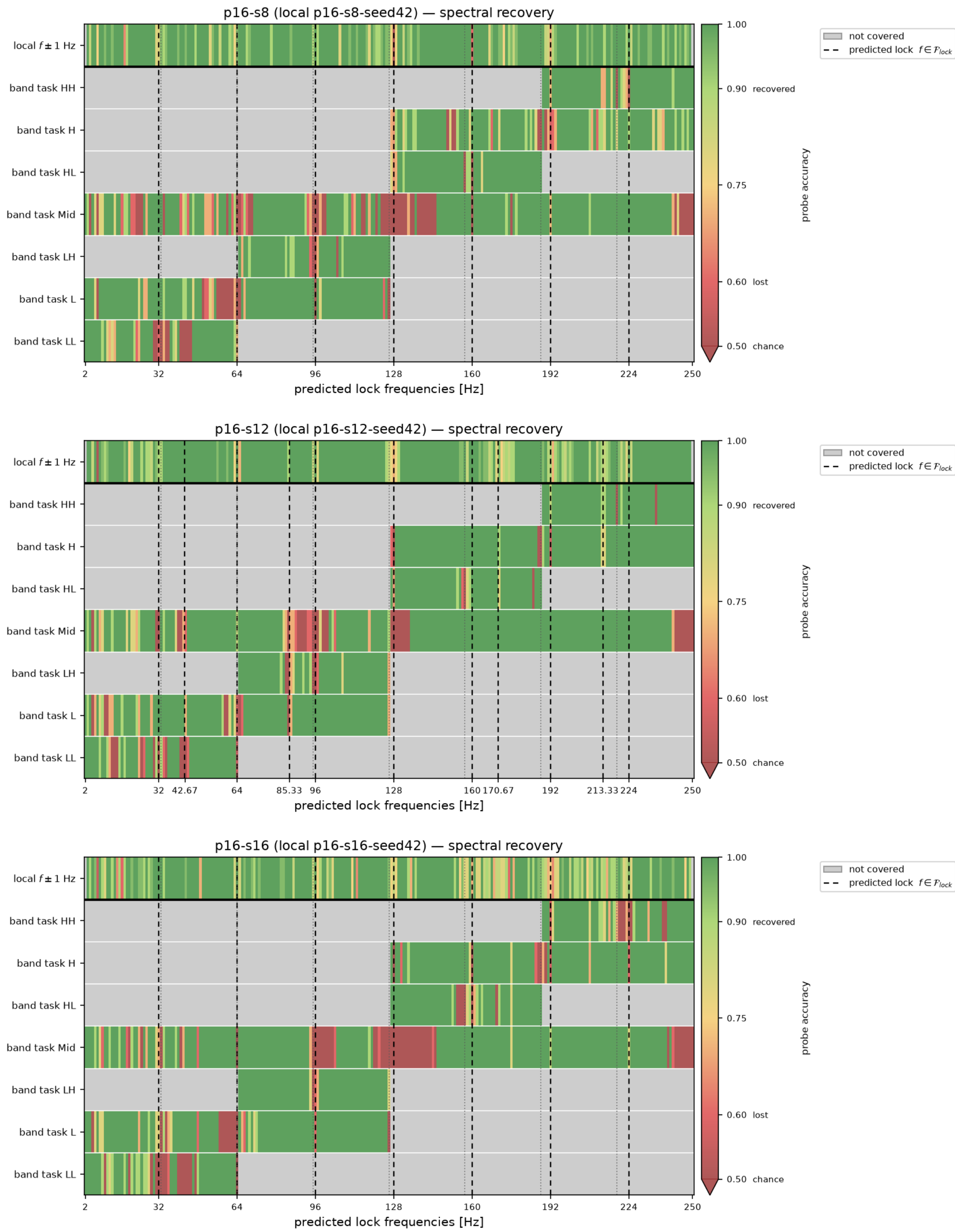

Figure 3: *Spectral recovery at $P = 16$, the stride lengthening from $S = 8$ to $S = 16$, so a feature that moves between panels belongs to the stride branch. Rows are the seven band tasks and, above the rule, the local $f \pm 1$ Hz task; colour is probe accuracy from chance to one, grey a frequency the task does not cover, dashed verticals $\mathcal{F}_{lock}$.*

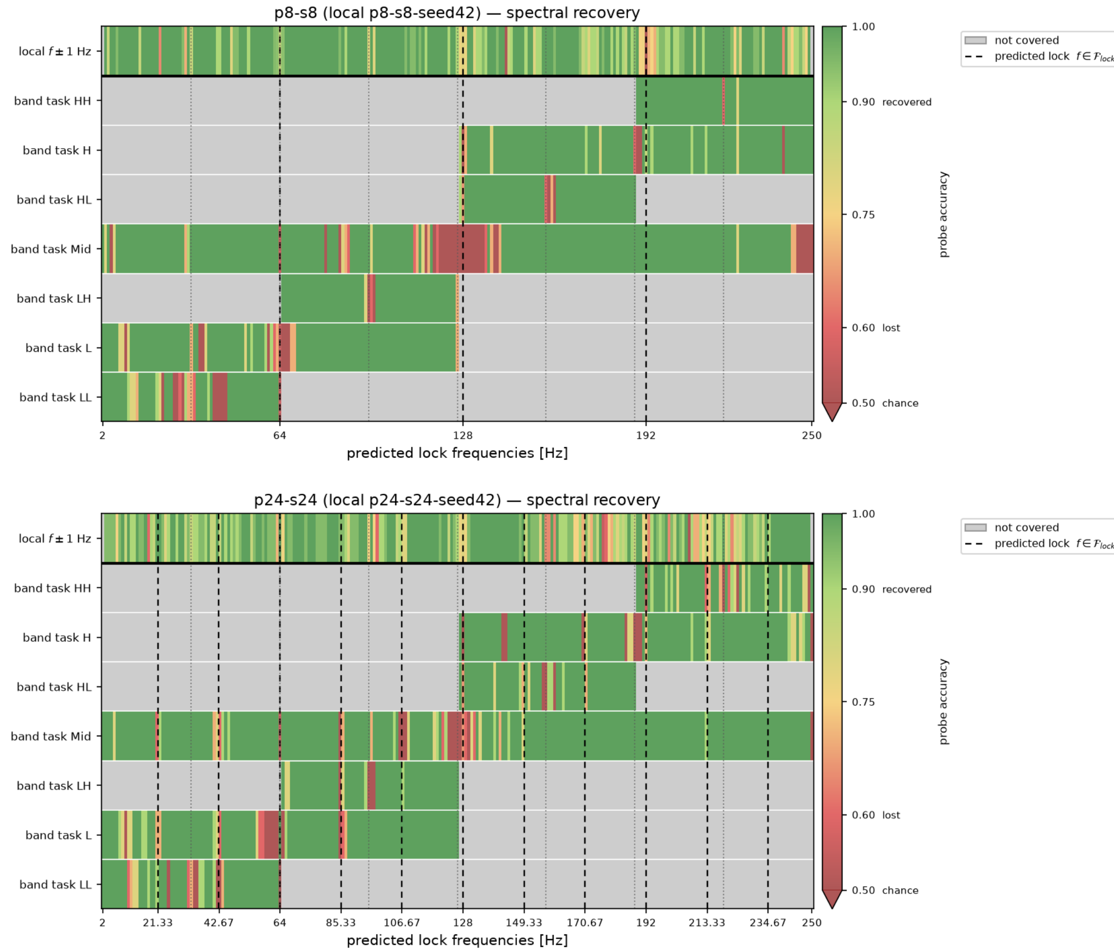

Figure 4: *Spectral recovery at the two $P = S$ geometries, where the patch length varies at zero overlap and the two branches coincide in both. Rows are the seven band tasks and, above the rule, the local $f \pm 1$ Hz task; colour is probe accuracy from chance to one, grey a frequency the task does not cover, dashed verticals $\mathcal{F}_{lock}$.*

The P=16, S=16 geometry is that of the published Chronos-Bolt checkpoint, but only the setting is shared: the model evaluated here is not that checkpoint, it is a variant retrained from scratch under the budget of this deliverable. Its reconstruction curve leaves the diagonal over wide stretches of the band. The figure does not license a claim about structural aliasing: it is behavioural, and records what the model outputs rather than what its representation retains.

Figure 2 repeats the reading on the other four geometries, and the pattern is the same in each of them: every model rebuilds some frequencies correctly while others are not rebuilt at all.

Spectral Recovery This reading is representational and records what the model still represents. The internal state is captured at the projected quantile head and a linear probe, standardisation then a principal component reduction then logistic regression, is scored by cross-validation. Two task families are run. The seven band tasks are those of Deliverable 1, each asking whether a tone falls in the lower or upper half of its band; their folds are grouped by frequency, so every phase of a held-out frequency is excluded from training and the probe has to generalise the boundary instead of memorising individual frequencies. An ungrouped split inflates those rows, so the earlier reading of them was optimistic. The local task poses one question of constant difficulty at every frequency, whether $f - 1$ Hz can be separated from $f + 1$ Hz, and holds out phases rather than frequencies; having no boundary artefact, it is the row the cross-geometry comparison uses. Cell colour is that cross-validated accuracy on a continuous scale from chance to one, so a partial loss is rendered as such rather than rounded into a bin; grey marks a frequency no task covers, which is a statement about the design and not about the model. Dashed markers on both panels are $\mathcal{F}_{lock}$ for that geometry.

Figure 3 and Figure 4 ask the same band of the representation instead of the forecast. In the local task the accuracy drops are narrow and fall close to some of the predicted lock frequencies; in the band tasks the drops instead extend over a range of frequencies around them.

These observations do not establish structural aliasing either, but they do show that the models represent certain frequencies with more difficulty than others. They are the empirical ground on which the hypotheses of Bayesian Analysis are built.

Limits of this design

The five runs were chosen for Deliverable 1 and they bound what can be concluded from the sweep above. Three of them have $P = S$, so their two branches coincide at every locked frequency and no observation in those geometries can be attributed to one branch. Stride-only sites exist in one geometry, P=16, S=12, and patch-only sites in two, so a claim about the stride branch rests on a single configuration and a claim about the patch branch on two. The configuration axes are equally short: the overlap ratio takes three values, and exactly one pair shares a stride while differing in patch size, which is the only place an effect of the patch can be separated from an effect of the stride. Every statement is therefore conditional on a narrow design.

To address this the models trained for the analysis are extended to the ones listed in Table 3. All the models are retrained from scratch under the same 100k-step budget and the same seed as the runs of Deliverable 1.

Run	**Patch (P)**	**Stride (S)**	**overlap**	**tokens**
1	8	8	0.00	256
2	16	8	0.50	255
3	16	12	0.25	170
4	16	15	0.06	136
5	16	16	0.00	128
6	24	8	0.67	254
7	24	12	0.50	169
8	24	15	0.38	135
9	24	16	0.33	127
10	24	20	0.17	102
11	24	24	0.00	85
12	32	8	0.75	253
13	32	12	0.62	169
14	32	15	0.53	135
15	32	16	0.50	127
16	32	20	0.38	101
17	32	24	0.25	85
18	32	28†	0.12	73
19	32	32	0.00	64

Table 3: *Every configuration published on the sweep repository, listed as in Deliverable 1: overlap, and the number of patch tokens at the* 2048*-sample training context. All are retrained from scratch under the same* 100*k-step budget and from the same seed, on the same GPU as the runs of Deliverable 1.* † *This stride does not divide the probing context, so the run can be swept but not fitted at it.*

Model	Claim	Estimand	Predicts	Input
A	$H1$ behavioural	$e^{\bar{\beta}}$: recovery at a lock / at its controls	< 1	`contrasts`
B	$H1$ representational	$e^{\theta_{lock}}$: codelength at a lock / elsewhere	> 1	`mdl_cells`
C	$H2$	σ_ϕ: spread of the deficit across phase	≈ 0	`contrasts`
D1	$H3$ location	θ_S, θ_P: dip depth on each grid, by LOO	both < 0	`collapse`
D2	$H3a$ stride branch	κ_S: measured / stride-predicted spacing	$= 1$	`sites`
D2	$H3b$ patch branch	κ_P: measured / patch-predicted spacing	$= 1$	`sites`
A′	$M1$ mitigation	δ_O: change in the deficit per unit of overlap	< 0	`contrasts`

Table 4: *The models, the quantity whose posterior decides each claim, the value the claim predicts, and the probing table each is fitted to. $H1$ is tested twice, on the forecast (A) and on the representation (B), so that a discrepancy between them stays visible. $H3$ is split into its two branches, since $\mathcal{F}_{lock}$ is a union and a frequency may belong to either. $M1$ is not a separate model but the configuration level of A, promoted to a named estimand.*

Bayesian Analysis

Deliverable 1 applied a two-part framework to $H1$ alone; this deliverable gives every hypothesis its own measurement, its own model and its own estimand. Table 4 lists the models with the value each claim predicts, Table 5 every prior, and Table 6 the rule that turns a posterior into a verdict. All are ordinary regressions; only the quantity on the left and the question the coefficient answers change between them.

Each model is built the same way: a measurement, a linear predictor μ_i that adds one offset per group the observation belongs to, and a likelihood. One coefficient is the estimand and the rest keep it unconfounded. Table 5 gives every prior with the role its parameter plays. The range over which the prior scale is varied as a sensitivity check is $\{0.25, 0.5, 1.0\}$; all are centred on no effect, so a posterior that has not moved from its prior is itself a report of an absent effect.

H1

The behavioural measurement is the forecast amplitude recovery $R = A_{\text{pred}}/A_{\text{true}}$, collected in matched triplets: each lock f_k with its two controls, all three sharing the same background realisation and the same phase, which is what makes the contrast paired. The contrast used is

$$d = \log(R_k + 0.01) - \tfrac{1}{2}\left[\log(R_- + 0.01) + \log(R_+ + 0.01)\right], \tag{1}$$

one number per triplet, negative when the lock recovers less than its neighbours. It can be formed only because the injected tone has a known frequency. Model A fits it as

$$d_i \sim \text{Student-}t_4(\mu_i,\ \sigma), \quad \mu_i = \beta_{c[i]} + u_{k[i]} + u_{b[i]}, \quad \beta_c \sim \mathcal{N}\Big(\bar{\beta} + \delta_O \widetilde{O}_c + \delta_P \widetilde{\log P}_c,\ \tau\Big). \tag{2}$$

Equation 2 has four components, each motivated separately.

- **Three offsets in μ_i**: β_c per configuration, u_k per lock harmonic and u_b per background draw, so that neither an unusual frequency nor a single corpus can be mistaken for the effect.
- **Student-t_4, not Gaussian.** Where the model rebuilds nothing, recovery is near zero and the logarithm returns very large contrasts; under a Gaussian a small number of them would determine the estimate.
- **β_c drawn around $\bar{\beta}$**, making the geometries instances of one phenomenon rather than unrelated experiments. Two consequences follow. $\bar{\beta}$ exists as a parameter, which $H1$ requires, being a claim about patch-based tokenisation and not about a single checkpoint; and each geometry contributes equal weight, so the largest design cannot dominate the largest effect when the number of usable triplets varies several-fold with the in-band lock count.

Model	Parameter	Prior	Role
A, C	$\bar{\beta}$	Student-$t_4(0, 0.5)$	population phase-lock effect, on the log-recovery scale
A	δ_O	Student-$t_4(0, 0.5)$	slope in $\widetilde{O}_c$, the centred overlap; one unit $= 0.5$ of O
A	δ_P	Student-$t_4(0, 0.5)$	slope in $\widetilde{\log P}_c$, the centred log patch size
A, C	$\tau, \sigma_k, \sigma_b, \sigma$	Half-Student-$t_4(0, 0.5)$	configuration, harmonic, background and residual scales
C	σ_ϕ	Half-$\mathcal{N}(0, 0.25)$	spread of the eight per-phase offsets
B	α_0	$\mathcal{N}(\log \bar{y}, 1)$	baseline log codelength
B	θ_{lock}	$\mathcal{N}(0, 0.5)$	log codelength expansion at a lock
B	k	Gamma$(2, 0.1)$	Gamma shape (dispersion)
B	$\sigma_{st}, \sigma_{geo}$	Half-$\mathcal{N}(0, 1)$	spread of the probe-stage and geometry offsets
D1	$\alpha_g, \theta_S, \theta_P$	$\mathcal{N}(0, 1)$	per-geometry off-grid level and the two grid labels
D1	σ	Half-$\mathcal{N}(0, 1)$	residual scale of $\log z_g(f)$
D2	κ_S, κ_P	$\mathcal{N}(0, 1)$	measured spacing per unit of predicted spacing
D2	σ_F	Half-$\mathcal{N}(0, 5)$ Hz	scatter beyond the sweep-resolution floor Δf

Table 5: *Every parameter with its prior and its role. The effect priors are those of Deliverable 1, extended by δ_P, which the design of Table 1 makes estimable.*

- **Two configuration-level covariates**, estimable together rather than confounded because the design varies overlap and patch size independently. Overlap enters as a ratio, patch size as $\log P$ because the patch grid has spacing f_s/P and equal steps in $\log P$ are equal ratios of spacing. The tilde denotes centring on the design, $\widetilde{O}_c = (O_c - \bar{O})/0.5$ and $\widetilde{\log P}_c = \log P_c - \overline{\log P}$, which leaves $\bar{\beta}$ the effect at the average configuration rather than an extrapolation to $O = 0$ and $P = 1$.

Model B poses the same question of the representation, on a frequency-local prequential codelength: at each f_c a probe separates tones at $f_c \pm 1\,\mathrm{Hz}$ from a captured internal state. Ten states are probed, the encoder [REG] token after each encoder block, the decoder state after each decoder block, the pre-projection vector and the quantile head; the decoder-side vectors are named decoder states because Chronos-Bolt carries [REG] in the encoder only. The regression is Gamma with a log link, so $e^{\theta_{lock}}$ is the factor by which the codelength of a locked frequency increases, with probe stage and geometry as zero-mean offsets since codelengths differ several-fold between stages. Deliverable 1's band tasks summarise hundreds of frequencies at once, so the lock indicator has nothing to attach to; they are kept as a cross-check.

H2

$H2$ uses the same triplets with one change. Equation 1 has already averaged over phase, so the phase index is kept instead and the response becomes the per-phase deficit $y_i = \log(R_{\mathrm{ctrl},i} + 0.01) - \log(R_{\mathrm{lock},i} + 0.01)$, which is d with its sign reversed and positive when the lock is worse. Model C is Model A on that response, without the configuration-level covariates and with the phase circle cut into eight slices, each given its own offset u_p:

$$y_i \sim \text{Student-}t_4(\mu_i,\ \sigma), \qquad \mu_i = \beta_{c[i]} + u_{k[i]} + u_{b[i]} + u_{p[i]}, \qquad u_p \sim \mathcal{N}(0, \sigma_\phi). \tag{3}$$

The estimand σ_ϕ is the spread of those eight offsets, and it answers the question with one number. If the deficit depends on where in its cycle the signal starts, the offsets differ and σ_ϕ is large. If the geometry alone determines it, they are equal and σ_ϕ is near zero, as $H2$ predicts. Slices are used rather than a fitted curve because they assume nothing about the form a dependence might take. Dropping u_p gives a nested null, and comparing the two by leave-one-out cross-validation (LOO) separates the size of the phase effect from the evidence that it exists.

H3

$H3$ was stated in Deliverable 1 as two claims, one per branch. Both are tested on the token collapse $z_g(f)$, the dispersion of the input-patch embeddings across patches, which is zero when consecutive patches are identical. All geometries share one sweep grid: a uniform 1 Hz sweep unioned with the predicted sites of every configuration. No geometry is therefore scored only where its own theory predicts a dip, and non-integer sites such as $42.\overline{6}$ Hz for S=12 are covered.

The first question is where the dips are located. Each swept frequency carries two labels, one per branch:

$$\log z_g(f) \sim \mathcal{N}(\alpha_g + \theta_S \cdot \mathrm{OnStrideGrid} + \theta_P \cdot \mathrm{OnPatchGrid},\ \sigma)\,. \tag{4}$$

The logarithm makes the labels multiplicative on the per-geometry off-grid level α_g. The two label coefficients θ_S and θ_P are the estimands, and $H3$ predicts both negative, so the two-label fit should win the LOO comparison against each label alone and against neither. The two labels are not competing explanations of the same dips, since a frequency may carry either or both. The two coefficients separate, however, only where exactly one label applies, which for θ_S means the geometries with $S \nmid P$.

The second question is whether the dips move. The fit above remains compatible with dips that merely sit on a fixed grid, so each branch is checked against its own scaling law. Every detected dip is assigned to the branch that predicts it, and sites belonging to both are set aside. The fundamental $\hat{f}^F_{1,g}$ of branch F in geometry g is then compared with the spacing that branch predicts, $\Delta^S_g = f_s/S_g$ or $\Delta^P_g = f_s/P_g$:

$$\hat{f}^F_{1,g} \sim \mathcal{N}\left(\kappa_F\,\Delta^F_g,\ \sqrt{\sigma_F^2 + \Delta f^2}\right). \tag{5}$$

The slope κ_F is the estimand, measured spacing per unit of predicted spacing, so $\kappa_F = 1$ denotes a branch that tracks its own arithmetic. The scatter σ_F about it is given a floor Δf, the sweep step, since a spacing read off a discrete grid is no more precise than one step; without the floor, sites landing exactly on the prediction drive σ_F to zero and the posterior degenerates.

H3a $H3a$ concerns the stride branch and predicts $\kappa_S = 1$: the stride sites track f_s/S one for one, so a site that remains fixed while S changes refutes it. It is identified by a series at fixed P, which the design supplies at P=16.

H3b $H3b$ concerns the patch branch and predicts $\kappa_P = 1$ in the same way. It is identified by pairs that share a stride and differ in patch size, which the design supplies at S=8.

Decisions and validation

Every threshold in Table 6 is fixed before a posterior is inspected, and validation comes before interpretation.

- **Prior predictive checks**, before collection.
- **Parameter recovery**: data simulated with a known effect, to confirm the model finds it. This is what separates "no effect" from "a design that cannot see one".
- **Posterior predictive checks**, stratified by frequency, phase, P, S and generator.
- **Model comparison** by LOO cross-validation, not by null-hypothesis testing.

Claim	Rule	Prior	Reads as
$H1$ supported	$\Pr(\bar{\beta} < \log 0.8 \mid D) \geq 0.95$	0.34	at least 20% attenuation at a lock
$H1$ refuted	$\Pr(\lvert\bar{\beta}\rvert < \log 1.1 \mid D) \geq 0.95$	0.14	any effect is smaller than 10%
$H2$ supported	$\Pr(\sigma_\phi < \log 1.1 \mid D) \geq 0.95$	0.30	phase moves the deficit by under 10%
$H3$ location	two-label fit wins LOO, $\theta_S, \theta_P < 0$	—	dips sit on both predicted grids
$H3a$, $H3b$	$\Pr(\lvert\kappa_F - 1\rvert < 0.1 \mid D) \geq 0.95$	0.05	that branch tracks its own parameter
$M1$ supported	$\Pr(\delta_O < 0 \mid D) \geq 0.95$, overlap term wins LOO	0.50	overlap reduces the deficit

Table 6: *Decision rules.* ***Prior*** *is the probability each rule already scored under the priors of Table 5: a posterior counts as evidence only in as much as it has moved away from that value. A claim is supported at* 0.95*, refuted at* 0.05*, and otherwise inconclusive.*

- **Sampling diagnostics**: NUTS (No-U-Turn Sampler) with four chains, and a fit reported only at $\hat{R} < 1.01$, effective sample sizes (ESS) above 1000 and no divergences. Where a model does not meet these thresholds, its estimand sign is cross-checked against the converged models and the frequentist probes; it is reported as provisional.

Semantic Layer

The ontology reproduced in Appendix A closes the loop from the signal to the decision. It represents the photovoltaic micro-grid, its telemetry and a single control agent, and it makes the tokenisation geometry a first-class object rather than a property of the model: a `pv:TimeSeriesSignal` has a `pv:PatchedRepresentation` carrying `pv:patchSize`, `pv:stride` and a derived `pv:overlapRatio`, and one signal may hold several, one per (P, S) under evaluation. Since the lock conditions are arithmetic on those same fields, the set of candidates is derivable from the knowledge base rather than hard-coded beside it, which is what lets an agent recompute it when a configuration or a sampling interval changes.

The point of contact with the Bayesian analysis is the following. An `pv:ObservabilityAssessment` does not derive a state from the geometry alone. It records the estimand, its effect size, its credible interval and its posterior probability, and assigns StructurallyAliased only to a frequency that is a lock candidate and also carries confirmed evidence of loss, $A = C \wedge B$ in the notation of Appendix A. Geometry alone yields a candidate, never a verdict. Because the confirmation threshold exceeds one half, confirmed loss and confirmed preservation cannot both hold. A frequency for which preservation is positively confirmed $(\neg A \wedge R)$ is assigned Observable; everything else is assigned PartiallyObservable: an unfinished test, a credible interval crossing the threshold, or conflicting metrics. That is the inconclusive verdict of Table 6 carried into the semantics, and its consequence is operational rather than documentary, because the assessment selects the control mode. Confirmed aliasing forces the safe mode and suspends forecast-driven battery decisions; a merely partially observable assessment reduces the battery cap and lowers decision confidence. Uncertainty about the representation is therefore propagated to the actuator instead of being resolved by assumption.

Mitigation Strategy

The mitigation carried through the design is increased patch overlap: shorter S at fixed P. Its mechanism is arithmetic: the stride branch has its lowest in-band member at f_s/S, so halving the stride doubles that frequency and thins the family. The same arithmetic bounds what it can achieve. Overlap does not affect the patch branch at all, so it thins at most one of the two and leaves the P/k-minute blind periods of the photovoltaic mapping in place, and at a fixed context the token count grows as $1/S$, which is a real constraint for an edge-deployed forecaster.

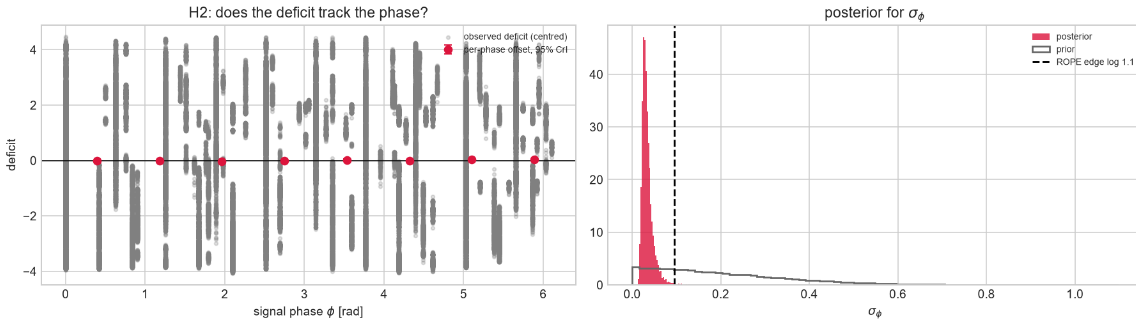

Figure 5: *Posterior for the phase term of $H2$. On the left the observed deficit against the signal phase ϕ, with the per-phase offset u_p and its 95% credible interval; on the right the posterior for σ_ϕ against its prior, the dashed line marking the ROPE edge at* $\log 1.1$. *Both panels pool every configuration fitted, so the estimate is not conditioned on a single geometry.*

Both halves are testable rather than asserted, and $M1$ in Table 6 is the test. Whether alternative mitigations can improve on this bound is discussed in the final deliverable.

Results

The frequency sweeps and the models of Bayesian Analysis point in the same direction: phase-locking affects the internal representation but does not appear to propagate to the forecast output.

On the behavioural side, locked frequencies are not systematically attenuated in the rollout. In several geometries the model reconstructs them as well as their unlocked neighbours, contrary to what the degeneracy alone would predict. The representational side points toward a different pattern: the prequential codelength at a locked frequency tends to be higher than at an unlocked one, suggesting that the internal state carries a measurable cost at those sites. This split, representation penalised and forecast intact, is consistent across generators and geometries and is the pattern the evidence points to.

The phase of the input signal does not appear to modulate the deficit appreciably, consistent with $H2$: the posterior for σ_ϕ in Figure 5 lies inside the region of practical equivalence (ROPE). It concentrates well below the threshold and is far tighter than the prior it started from, so the data and not the prior place it there, and the eight per-phase offsets u_p are not separable from one another. The size of the deficit therefore follows from the geometry and from the frequency, and not from where in its cycle the signal begins, which is what licenses the remaining models to average over phase and which also removes phase randomisation from the mitigations available to an operator.

The token-collapse dips align with the stride grid and shift when the stride changes, consistent with $H3a$, and Figure 6 shows the pattern holding across configurations and generators; the patch branch is not separately identifiable in the current design.

The overlap-based mitigation of Mitigation Strategy is left to the final deliverable, where the representational cost is what it will be measured against.

Models A and C have not reached the preregistered convergence thresholds; their estimand signs are nonetheless consistent with the converged representational model (B) and with the independent frequentist sweeps, so the qualitative conclusions hold provisionally. The final deliverable will either report consolidated posteriors or, if convergence remains out of reach, document the limitation and fall back on the frequentist evidence.

Figure 6: *Collapse profiles against the two candidate grids. Rows are the fitted configurations and columns the three generators; each panel plots the profile over the sweep band, with the stride grid cf_s/S dashed and the patch grid kf_s/P dotted. A dip that sits on the dashed grid and follows it as the stride changes belongs to the stride branch.*

Security Considerations

The forecaster is not a standalone artefact. It sits inside a control loop in which its output drives battery dispatch and curtailment on a photovoltaic micro-grid, so a forecasting failure can become a control failure, and the property under test in this deliverable has to be read as a security property and not only as an accuracy one. The NIST AI Risk Management Framework [1] is the reference we adopt for that reading, because it separates the characteristics we can evidence here, validity and reliability, security and resilience, from the governance functions that turn evidence into a control, and because it asks for known limitations to be mapped and measured rather than merely disclosed.

The limitation itself is unusual, and characterising it correctly determines what follows. Phase-locking is deterministic, attacker-independent and present at deployment time. It is not introduced by an adversary and it is not a defect in the training data; it is a consequence of the tokenisation geometry, and it is closed-form. The set of locked frequencies follows from (f_s, P, S) alone, and those three numbers are configuration, not weights. Structural aliasing is the phenomenon that may then appear in the learned representation, and phase-locking is necessary for it but not sufficient, so the locked set bounds where the model can be blind while Bayesian Analysis is what establishes where it is. An adversary therefore needs no model access, no gradients, no query budget and no knowledge of the training corpus to enumerate that set, and only observation of the deployed forecaster to find which of its members carry a deficit. In the taxonomy of adversarial machine learning [2] this places the weakness with the availability violations rather than with integrity or privacy, and it places the attack in the white-box-free category, which is what a conventional threat model tends to miss because it reasons about perturbations of the input tensor rather than of the physical process.

The attack surface follows directly. The perturbation an adversary needs is physical and small: a periodic component injected at, or dithered onto, an aliased frequency by any controllable load, inverter switching pattern or curtailment schedule is present in the plant and absent from the forecast, because there the tokenisation maps it to the same latent state as its absence. Two consequences are operationally significant. The signal is invisible rather than misread, so residual-based anomaly detection downstream of the forecaster does not see it either; and the same arithmetic that makes the exploit inexpensive also makes it stealthy, since the injected component is legitimate plant behaviour at a frequency the operator has no reason to treat as special. Detection has to be placed before tokenisation, on the raw telemetry, or it is placed inside the blind spot it is meant to guard.

What the framework then requires is that this be mapped, measured and managed rather than merely noted. Table 3 and the lock arithmetic map it, since together they enumerate the candidate set of every deployed configuration. Bayesian Analysis measures it, and the prior probability recorded beside each threshold keeps the reading auditable rather than asserted. Management is twofold: in the model the overlap mitigation thins one branch and leaves the other, so residual risk survives by construction; in the system the semantic layer degrades actuation through a PartiallyObservable assessment instead of failing silently, which is the resilience the framework asks for.

A weakness becomes a vulnerability only when someone can reach it, and the final deliverable will make that step explicit. The five elements it needs are already determined by what precedes.

- **Adversary capabilities.** No model access of any kind, and no knowledge of the training corpus; only the ability to drive a periodic load on the plant, which an inverter switching schedule or a switchable load already provides. With a published Chronos-Bolt checkpoint the assumption is weaker still, since its training corpus is public and the adversary holds it whether or not the attack requires it.
- **Attack surface.** The telemetry stream upstream of tokenisation, reached by acting on the physical process rather than on the data path.
- **Manipulation of telemetry.** A small periodic component at an aliased frequency, which is legitimate plant behaviour and survives any plausibility check on the raw signal.
- **Operational consequence.** The component is present in the plant and absent from the forecast, so

battery dispatch and curtailment are decided on a signal the plant is not producing, and no residual-based detector downstream flags the discrepancy.

- **Residual risk.** Overlap thins the stride branch alone, so the patch branch stays exploitable at every deployed configuration and cannot be closed by the mitigation carried here.

Appendix A: Ontology of the Photovoltaic Application Domain

This document defines a compact ontology for a photovoltaic system governed by one control agent. It connects environmental and electrical observations to anomaly recognition, Chronos-style patched representations, structural-aliasing evidence, and bounded power-routing decisions [3]. Routing is included only as the operational endpoint of the semantic model; the principal concern remains whether the representation preserves the information on which the agent relies.

Domain Description

The modeled system contains four energy assets: a photovoltaic array, battery storage, an aggregate internal load, and an external grid. The internal load is a passive sink representing a factory or a fleet of machines. Its demand is observed, but it has no agent, machine-level taxonomy, curtailment command, or scheduling model. The external grid is an available bidirectional boundary bus. It supplies residual demand and accepts residual production.

Exactly one `pv:ControlAgent` coordinates the four passive assets. It observes one synchronized control cycle, assigns environmental, anomaly, and observability states, selects a control mode, and emits one or more directional decisions. There is no peer-to-peer negotiation among assets.

The seven admissible transfer directions are

$$\begin{aligned} &\text{PV} \to \text{Load}, && \text{PV} \to \text{Battery}, && \text{PV} \to \text{Grid}, \\ &\text{Battery} \to \text{Load}, && \text{Battery} \to \text{Grid}, && \text{Grid} \to \text{Load}, \\ &&& \text{Grid} \to \text{Battery}. && \end{aligned} \tag{6}$$

They satisfy the cycle-level balance

$$P_{\text{pv}} + P_{\text{bat,dis}} + P_{\text{grid,in}} = P_{\text{load}} + P_{\text{bat,ch}} + P_{\text{grid,out}}. \tag{7}$$

All terms denote non-negative magnitudes; import and export are therefore kept separate rather than encoded by an ambiguous sign.

Purpose and Scope

The ontology provides the smallest semantic layer needed to represent the physical boundary, validated observations, environmental context, three selected anomalies, and the effect of patching on model observability. It does not model individual machines, market prices, dispatch optimisation, grid faults, or a deployed real-time controller. Load demand, battery SoC, asset limits, and the optional grid-exchange reference are asserted or simulated because they are not present in the photovoltaic telemetry dataset.

The environmental states are operational generation proxies, not meteorological diagnoses. Likewise, the routing rules describe a deterministic conceptual policy rather than the central experimental contribution. Structural aliasing remains an empirically tested property of a signal-representation pair; no fixed synthetic frequency is transferred to the one-minute photovoltaic signal.

Ontology Representation

Core classes and closed vocabularies Table 7 defines the compact class interface. The four asset subclasses form a disjoint union; each `pv:PVSystem` contains exactly one asset of each type and is managed by exactly one control agent. Closed operational vocabularies are represented as named individuals rather than deeper class hierarchies.

Class	**Role**
`pv:PVSystem`	Aggregate system and scope of each control cycle.
`pv:EnergyAsset`	Parent of the four passive physical assets.
`pv:PhotovoltaicArray`	Intermittent renewable source measured through PV telemetry.
`pv:BatteryStorage`	Bounded storage asset that can charge, discharge, or hold.
`pv:InternalLoad`	Aggregate passive demand of a factory or machine fleet.
`pv:ExternalGrid`	Reliable bidirectional boundary bus for residual balancing.
`pv:ControlAgent`	The single decision-making entity.
`pv:SystemObservation`	Time-stamped snapshot of PV, environment, battery, load, and grid reference.
`pv:TimeSeriesSignal`	Physical signal and its temporal descriptors.
`pv:PatchedRepresentation`	A model input representation defined by patch and stride.
`pv:ObservabilityAssessment`	Evidence linking a signal-representation pair to an observability state.
`pv:GenerationContext`	Operational proxy for expected solar generation.
`pv:AnomalyState`	Detected physical or sensing abnormality.
`pv:ObservabilityState`	Semantic assessment of information retained after patching.
`pv:ControlMode`	Normal, conservative, or safe operating policy.
`pv:PowerRoute`	Closed vocabulary of the seven admissible source–destination pairs.
`pv:PowerRoutingDecision`	One directional transfer or hold decision with an optional bounded setpoint.

Table 7: *Core classes in the simplified ontology.*

The closed vocabularies are:

Generation context: ClearSky, PartialCloud, Overcast, and Nighttime.
Anomaly: InverterTrip, RampEvent, and SensorMalfunction.
Observability: Observable, PartiallyObservable, and StructurallyAliased.
Control mode: NormalMode, ConservativeMode, and SafeMode.
Power route: PVToInternalLoad, PVToBattery, and PVToExternalGrid;
BatteryToInternalLoad and BatteryToExternalGrid;
ExternalGridToInternalLoad and ExternalGridToBattery.

Formally, each vocabulary class is an OWL-style enumeration of the listed named individuals (`oneOf`). The five vocabulary classes are pairwise disjoint, and all listed values are declared globally different (`AllDifferent`). This prevents a value such as SafeMode from being silently identified with ClearSky or NormalMode under open-world semantics. Each valid time-stamped observation has exactly one generation context. Each completed observability assessment has exactly one observability state, and each routing decision records exactly one control mode. Co-issued decisions based on the same synchronized observation must record the same mode. Anomalies are not declared mutually exclusive because, for example, an abrupt inverter trip also creates a large power change. Deterministic control is obtained through the priority rules in the closed-loop section.

Object properties The semantic relations in Table 8 connect the physical boundary to the inference and control layers. Each active `pv:PowerRoutingDecision` has exactly one route value, one source, and one distinct destination. Its non-negative setpoint magnitude is optional because `Balance` may denote passive residual exchange. A cycle that simultaneously serves the load, charges the battery, and exports a residual uses three instances of the same decision class, all based on the same observation and mode.

Property	**Domain**	**Range or meaning**
`pv:hasAsset`	`pv:PVSystem`	`pv:EnergyAsset`
`pv:managedBy`	`pv:PVSystem`	exactly one `pv:ControlAgent`
`pv:hasObservation`	`pv:PVSystem`	`pv:SystemObservation`
`pv:hasSignal`	`pv:PhotovoltaicArray`	`pv:TimeSeriesSignal`
`pv:hasRepresentation`	`pv:TimeSeriesSignal`	`pv:PatchedRepresentation`
`pv:assessesRepresentation`	`pv:ObservabilityAssessment`	exactly one `pv:PatchedRepresentation`
`pv:hasGenerationContext`	`pv:SystemObservation`	zero or one `pv:GenerationContext` value
`pv:hasAnomaly`	`pv:SystemObservation`	zero or more `pv:AnomalyState` values
`pv:hasObservabilityState`	`pv:ObservabilityAssessment`	one `pv:ObservabilityState` value
`pv:selectsMode`	`pv:PowerRoutingDecision`	exactly one `pv:ControlMode` value
`pv:makesDecision`	`pv:ControlAgent`	`pv:PowerRoutingDecision`
`pv:basedOn`	`pv:PowerRoutingDecision`	observation, assessment, or inferred state
`pv:hasRoute`	`pv:PowerRoutingDecision`	zero or one `pv:PowerRoute` value
`pv:hasSourceAsset`	`pv:PowerRoutingDecision`	zero or one `pv:EnergyAsset`
`pv:hasDestinationAsset`	`pv:PowerRoutingDecision`	zero or one `pv:EnergyAsset`

Table 8: *Object-property signatures.*

`pv:hasRepresentation` is intentionally non-functional because one signal can be evaluated under several (P, S) configurations. Each assessment identifies exactly which representation it evaluates. Each decision is based on exactly one current observation and its completed assessment; its own `pv:selectsMode` value therefore scopes the mode without accumulating incompatible modes on the persistent agent. The range of `pv:basedOn` is the union of observation and inferred-assessment entities, rather than the intersection that multiple RDFS range declarations would imply.

OWL's open-world semantics do not by themselves validate that endpoint types match a named route. The operational validator therefore admits only the seven ordered pairs in Equation 6, rejects identical endpoints, and checks consistency among `pv:hasRoute`, source, and destination. A hold decision has no route or endpoints and has a zero or absent setpoint. The `pv:basedOn` relation is retained only as the minimum evidence link required to justify a selected mode; no separate trace model is introduced.

Data properties and units The synchronized observation combines the measured photovoltaic and environmental variables with asserted or simulated battery, load, and grid-reference values. The control-critical quantities are PV power P_{pv}, tilted irradiance G_{tilt}, normalized battery state of charge $s \in [0, 1]$, and aggregate load demand $P_{\mathrm{load}} \geq 0$. Each has a separate validity flag derived from presence, finiteness, freshness, and a configured physical range. Horizontal irradiance, air temperature, wind speed, and wind direction remain environmental evidence but do not independently block the controller. The signed `pv:gridExchangeReferenceW` is optional, with positive values denoting export; it is usable only when present, finite, and fresh.

Missing or stale data cannot be inferred safely from OWL's open-world assumption alone. An operational validation layer therefore compares the observation time stamp with a configurable staleness limit, validates numeric ranges, and asserts the four critical Boolean flags consumed by the ontology. This keeps data-quality validation distinct from semantic classification.

Owner	Data properties
`pv:SystemObservation`	`pv:pvPowerW`, `pv:powerRampRateWPerS`, `pv:loadDemandW`, optional `pv:gridExchangeReferenceW`, `pv:tiltedIrradianceWm2`, `pv:horizontalIrradianceWm2`, `pv:airTemperatureC`, `pv:windSpeedMs`, `pv:windDirectionDeg`, `pv:batterySoC`, `pv:observationTimestamp`, `pv:pvPowerValid`, `pv:tiltedIrradianceValid`, `pv:batterySoCValid`, and `pv:loadDemandValid`.
`pv:BatteryStorage`	`pv:capacityWh`, `pv:reserveSoC`, `pv:targetSoC`, `pv:maximumSoC`, `pv:maximumChargePowerW`, and `pv:maximumDischargePowerW`.
`pv:TimeSeriesSignal`	`pv:signalFrequencyHz`, `pv:samplingFrequencyHz`, `pv:periodSamples`, `pv:amplitudeW`, `pv:timeSpanSeconds`, and `pv:phaseOffsetSamples`.
`pv:PatchedRepresentation`	`pv:patchSize`, `pv:stride`, and derived `pv:overlapRatio`.
`pv:ObservabilityAssessment`	`pv:analysisStatus`, `pv:assessmentMethod`, `pv:assessmentMetric`, `pv:strideLockingRatio`, `pv:patchLockingRatio`, `pv:lockingTolerance`, `pv:effectEstimate`, `pv:posteriorLossProbability`, `pv:credibleIntervalLower`, `pv:credibleIntervalUpper`, and `pv:assessmentTimestamp`.
`pv:PowerRoutingDecision`	optional non-negative `pv:powerSetpointW`, `pv:decisionConfidence`, `pv:batteryCommand` in {Charge, Discharge, Hold}, `pv:pvCommand` in {Connect, Isolate}, and `pv:gridCommand` in {Import, Export, Balance}.

Table 9: *Quantitative property groups and units encoded by their names.*

A valid patched representation satisfies $P \geq 1$ and $1 \leq S \leq P$. Overlap is descriptive metadata:

$$\gamma = \frac{P - S}{P}. \tag{8}$$

It is not used as a standalone rule for declaring aliasing.

Inference Rules

Environmental generation context Let $g_0 < g_1 < g_2$ be site-configurable tilted-irradiance thresholds and let G_t denote the validated value of G_{tilt} at time t. The active generation context is

$$C_G(G_t) = \begin{cases} \mathsf{Nighttime}, & G_t \leq g_0, \\ \mathsf{Overcast}, & g_0 < G_t < g_1, \\ \mathsf{PartialCloud}, & g_1 \leq G_t < g_2, \\ \mathsf{ClearSky}, & G_t \geq g_2. \end{cases} \tag{9}$$

Values $g_0 = 0$, $g_1 = 150$, and $g_2 = 800\ \mathrm{W\,m^{-2}}$ are retained only as an illustrative configuration compatible with the current coursework. They are not universal meteorological boundaries. In particular, low irradiance can also be caused by solar position; the four values must therefore be read as generation-context proxies rather than certain weather diagnoses.

Anomaly detection and precedence Let $v_P(t)$, $v_G(t)$, $v_s(t)$, and $v_L(t)$ denote the validity of PV power, tilted irradiance, battery SoC, and aggregate load demand. The control-critical validity predicate is

$$V_t = v_P(t) \wedge v_G(t) \wedge v_s(t) \wedge v_L(t). \tag{10}$$

If V_t is false, no physical anomaly or forecast-driven route is inferred from that observation. A valid irradiance value may still support a descriptive generation context, but the operational validator asserts SensorMalfunction and the cycle proceeds directly to safe-mode selection. Within a validated cycle, absence of the three listed anomaly values is interpreted locally as “none detected”; this is a scoped closed-world policy, not a global OWL assumption. The signed ramp rate, measured in watts per second when Δt is expressed in seconds, is

$$r_t = \frac{P_t - P_{t-1}}{\Delta t}. \tag{11}$$

The anomaly rules are expressed with configurable thresholds: g_{trip} for sufficient irradiance, p_0 for near-zero power, and $r_{\max}$ for maximum admissible ramp magnitude.

$$\neg V_t \longrightarrow \textsf{SensorMalfunction}, \tag{12}$$

$$V_t \wedge G_t \geq g_{\text{trip}} \wedge |P_t| \leq p_0 \longrightarrow \textsf{InverterTrip}, \tag{13}$$

$$V_t \wedge V_{t-1} \wedge |r_t| > r_{\max} \longrightarrow \textsf{RampEvent}. \tag{14}$$

No numerical value is inherited for $r_{\max}$ because it must be calibrated after invalid values and outliers have been excluded. The controller evaluates sensor validity first. A sensor malfunction therefore cannot be reinterpreted as a physical ramp or inverter trip. If a valid inverter trip also satisfies the ramp predicate, the safe response to the trip takes precedence over ramp smoothing.

Signal observability and structural aliasing For a signal frequency f, sampling frequency f_s, patch size P, and stride S, define a stride-lock ratio and a patch-periodicity ratio

$$\rho_S = \frac{fS}{f_s}, \qquad \rho_P = \frac{fP}{f_s}. \tag{15}$$

A stride phase-lock candidate L_S and a within-patch repetition candidate L_P are

$$\begin{aligned} L_S &= \left[\exists m \in \mathbb{Z}^+ : |\rho_S - m| \leq \varepsilon_S\right], \\ L_P &= \left[\exists k \in \mathbb{Z}^+ : |\rho_P - k| \leq \varepsilon_P\right], \\ C &= (L_S \vee L_P) \wedge \left(0 < f < \frac{f_s}{2}\right), \end{aligned} \tag{16}$$

where ε_S and ε_P are tied to frequency resolution and are never hardcoded frequency bands. Only L_S expresses equality of consecutive patch phases; L_P records whole-cycle repetition inside a patch and is retained as an H3 experimental factor. Their union C is a candidate geometry class, not proof that two domain concepts are indistinguishable.

This is a division of labour, and it bounds what the ontology can conclude on its own. Phase-locking is *derivable*: L_S, L_P and C follow by rule from `pv:signalFrequencyHz`, `pv:samplingFrequencyHz`, `pv:patchSize` and `pv:stride`, all of which the knowledge base already holds, so a reasoner can enumerate the candidate set for any configuration without observing a model. Structural aliasing is *not* derivable and is only recorded: it enters through B below, which is posterior evidence from an experiment external to the ontology, and no rule here can produce it. An agent using this ontology therefore detects phase-locking and reports structural aliasing.

Let q denote the task-relevant information being tested, for example frequency or amplitude. Let $\Delta_{\text{loss}}(q)$ be its predeclared behavioral or representational loss relative to matched controls, δ the maximum practically negligible loss, D the experimental evidence, and $\tau > 1/2$ the posterior confidence threshold. Define confirmed information loss B and confirmed preservation R as

$$B = \left[\Pr(\Delta_{\text{loss}}(q) > \delta \mid D) \geq \tau\right], \qquad R = \left[\Pr(\Delta_{\text{loss}}(q) \leq \delta \mid D) \geq \tau\right]. \tag{17}$$

Because $\tau > 1/2$, B and R cannot both hold for the same assessment. The assessment interface is designed to receive the planned offline evidence already specified in Deliverable 1: a local amplitude-recovery contrast with a heavy-tailed Student-t_4 likelihood and a Gamma regression with log link for non-negative prequential-MDL codelength. The population contrast uses the weakly informative prior $\bar{\beta} \sim$ Student-$t_4(0, 0.5)$; the Gamma model uses $\theta_{\text{lock}} \sim \mathcal{N}(0, 0.5)$ and shape $k \sim \text{Gamma}(2, 0.1)$. The stored record identifies the tested (P, S) configuration, matched controls, method and metric, effect estimate, 95% credible interval, and posterior probability of at least 20% attenuation. These are evidence fields, not universal ontology thresholds. Until that planned inference has completed and satisfies the predeclared criterion, no instance may be asserted as StructurallyAliased.

The three observability values are assigned as follows:

$$C_O = \begin{cases} \textsf{StructurallyAliased}, & A, \\ \textsf{Observable}, & \neg A \wedge R, \\ \textsf{PartiallyObservable}, & \neg A \wedge \neg R, \end{cases} \qquad A = C \wedge B. \tag{18}$$

Observable therefore means that positive evidence shows the tested representation preserves the task-relevant information within tolerance δ. It does not mean global observability of the physical system, nor does it merely mean that aliasing was not found. Candidate geometry may still be Observable when the experiment demonstrates preservation. PartiallyObservable is the epistemic fallback: preservation has not been certified, while structural aliasing has not been confirmed. It includes a candidate with inconclusive evidence, an unfinished test, a credible interval crossing the decision threshold, conflicting metrics, or no preservation evidence. The term means *partially trusted*, not that a fixed fraction of samples or sensors is visible.

H1 compares each candidate frequency with matched nearby non-candidate controls and predicts higher MDL codelength or lower amplitude recovery; absent or non-localized loss refutes H1. H2 tests whether the loss at a candidate frequency remains approximately stable across sampled phase offsets; systematic phase dependence refutes that hypothesis. H3 tests whether candidate degradation sites move proportionally to $1/S$ or $1/P$ when the corresponding parameter varies. These hypotheses contribute controlled evidence to D, but remain falsifiable experimental factors rather than ontology subclasses or automatic action rules. Sensor validity is modeled separately as an anomaly and never relabels a model representation as structurally aliased.

When an experimental generator provides a phase ϕ in radians, the corresponding temporal offset is converted before storage:

$$o_{\text{samples}} = \frac{\phi f_s}{2\pi f}. \tag{19}$$

Neither ϕ nor o_{samples} is compared directly with stride to infer an anomaly.

The semantic propagation path is explicit:

$$\begin{aligned} \texttt{pv:TimeSeriesSignal} &\longrightarrow \texttt{pv:PatchedRepresentation} \\ &\longrightarrow \texttt{pv:ObservabilityAssessment} \\ &\longrightarrow \texttt{pv:ControlMode} \longrightarrow \texttt{pv:PowerRoutingDecision}. \end{aligned} \tag{20}$$

Aliasing can therefore reduce confidence in forecast-derived production, ramp expectations, forecast-driven anomaly recognition, and routing. It does not alter the directly observed values of G_{tilt}, P_{pv}, load demand, or SoC; those values are governed by the separate validity path.

Closed-Loop Control

Mode selection The agent evaluates each synchronized control cycle using the strict priority

$$\textsf{SafeMode} \succ \textsf{ConservativeMode} \succ \text{RampOverride} \succ \text{NormalRouting}. \tag{21}$$

Let A_{safe} mean that SensorMalfunction or InverterTrip is present. The selected mode is

$$M_t = \begin{cases} \textsf{SafeMode}, & A_{\text{safe}} \vee (C_O = \textsf{StructurallyAliased}), \\ \textsf{ConservativeMode}, & C_O = \textsf{PartiallyObservable}, \\ \textsf{NormalMode}, & \text{otherwise}. \end{cases} \tag{22}$$

In SafeMode, the battery command is Hold and the grid balances the internal load. The PV command is Isolate only for an inverter trip; for sensor malfunction or confirmed structural aliasing it remains connected, but forecast- and ramp-driven battery decisions are suspended. When load demand itself is invalid, Balance carries no invented numerical setpoint. In ConservativeMode, the same seven route types remain available with a configurable reduced battery-power cap, normal target/reserve bounds, and low decision confidence. A concurrent ramp cannot activate either extended ramp bound; this implements the precedence in Equation 21.

Routing policy The ordinary policy serves the internal load before charging the battery or exporting energy. Let

$$P_{\text{sur}} = (P_{\text{pv}} - P_{\text{load}})^{+}, \qquad P_{\text{def}} = (P_{\text{load}} - P_{\text{pv}})^{+}. \tag{23}$$

Here *daylight* abbreviates {ClearSky, PartialCloud, Overcast}. A cycle issues a route only for a strictly positive transfer and may issue several co-mode decisions. All setpoints are clipped to SoC, capacity, and charge/discharge limits.

Figure 7 shows the priority scan and Table 10 states its boundary cases. The conservative branch rejoins the ordinary load-first router under its reduced cap. The ramp branch is available only in NormalMode.

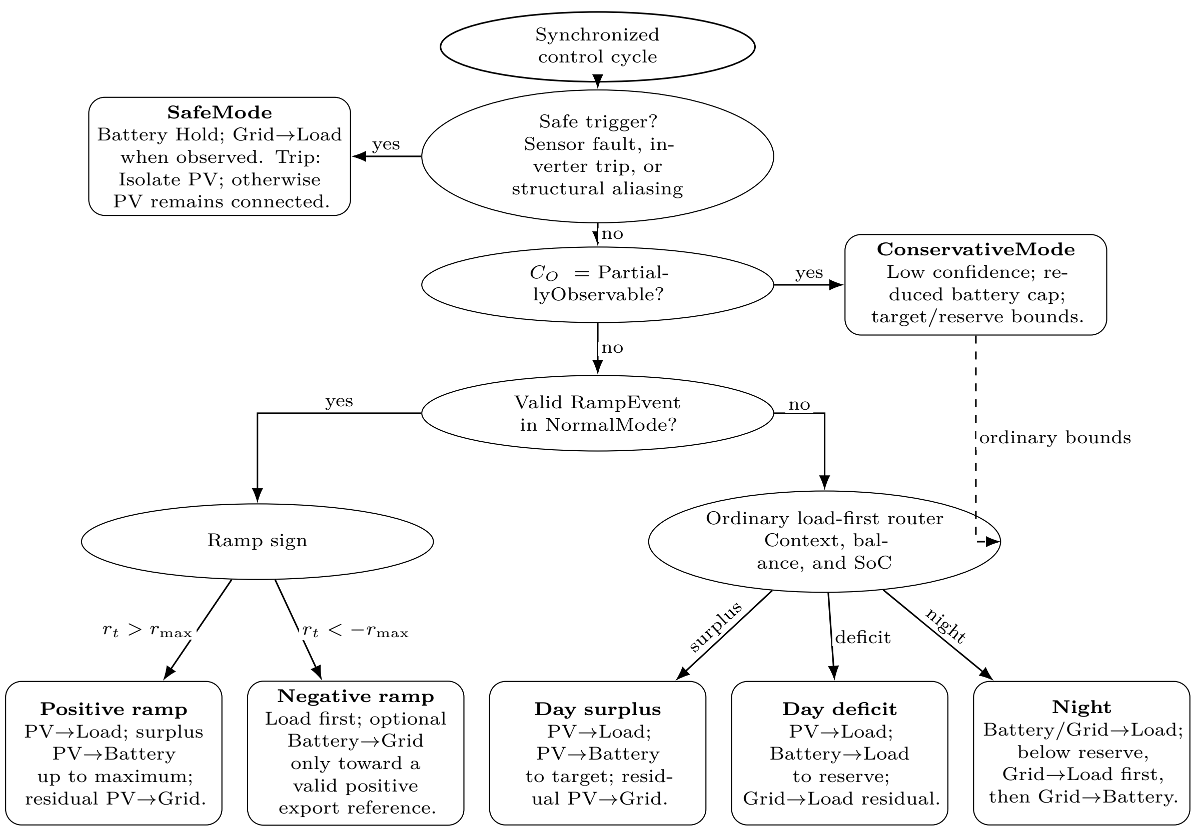

Figure 7: *Load-first routing decision tree. Elliptical bubbles are ordered gates; rounded bubbles are terminal policies. Safe and conservative decisions precede the normal-mode ramp override, while the ordinary router covers daylight surplus, daylight deficit, and nighttime operation.*

Condition	**Ordered route set**	**Constraint**
Daylight surplus	PV→Load; PV→Battery; PV→Grid	Serve demand, charge to target, then export the residual.
Daylight deficit	PV→Load; Battery→Load; Grid→Load	Discharge only to reserve; the grid covers the remaining deficit.
Night, $s > s_{\text{reserve}}$	Battery→Load; Grid→Load	Battery serves demand to reserve; the grid supplies the residual.
Night, $s \leq s_{\text{reserve}}$	Grid→Load; if $s < s_{\text{reserve}}$, Grid→Battery	Supply the load first; charge only below reserve and stop at reserve.
Positive ramp with surplus	PV→Load; PV→Battery; PV→Grid	NormalMode may charge beyond target, never beyond maximum SoC.
Negative ramp with valid positive reference	load-serving routes; Battery→Grid	Export only after serving the load, within residual discharge capacity and above reserve.
Negative ramp without usable reference	ordinary surplus/deficit routes	Missing, stale, non-finite, zero, or negative reference disables export catch-up.
ConservativeMode	ordinary load-first routes	Reduced battery cap; target/reserve bounds; no ramp extensions.
SafeMode	Grid→Load when observed	Battery Hold; grid Balance; isolate PV only for inverter trip.

Table 10: *Routing matrix for the seven admissible transfer directions.*

For a negative ramp, Battery→Grid is admissible only after the internal load has been fully served, the export reference is present, fresh, finite, and positive, measured export is below that reference, the battery remains above reserve, and discharge capacity remains after load support. If any condition fails, the ordinary load-first route applies. At night, Grid→Battery begins only after Grid→Load has been satisfied and stops when reserve is restored. These rules preserve Equation 7 without turning the routing policy into a separate optimisation problem.

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